

Alejandro Soler González

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Physicist (M.Sc. Space Systems Engineering, **GPA 9.1/10**) moving into **quantitative research & trading**. Built and ran a **live** cross-venue execution & data-capture system in **Rust** on a market-microstructure edge, validated on 1,350 real settlements and published as an SSRN research note. Available September 2026.

EDUCATION

M.Sc. in Space Systems Engineering — Universidad Europea, Madrid 2025 – Sep 2026 (expected)
GPA 9.1/10

- **Master's Thesis** (in progress): real-time trajectory optimization for autonomous rocket landing — posed the final 2 km of powered descent as a sequence of **convex programs (SOCP)** solved in closed loop inside a self-built **3-/6-DoF simulator** — the same convex-guidance method family used for SpaceX booster landings. [Interactive demo](#).

B.Sc. in Physics — Universidad Complutense de Madrid (bilingual Spanish/English program) 2021 – 2025

- **Bachelor's Thesis (9.2/10)**: multi-agent **Bayesian search** — coordinated agents locate the maximum of an unknown field via **Gaussian-process (Kriging) regression**, balancing exploration vs. exploitation under uncertainty. Selected for Docta Complutense (UCM's distinguished-thesis repository). [Interactive demo](#).
- Coursework: mathematical methods, statistical physics, scientific computing, probability & statistics; quantum-computing program (3 algorithms implemented in **Qiskit**).

EXPERIENCE & PROJECTS

Quantitative Research — Cross-Venue BTC Oracle Basis (Polymarket / Kalshi) Feb – May 2026

- Discovered a **systematic, signed cross-venue mispricing** in hourly Bitcoin prediction markets: the two venues settle on **different oracles** (Binance USDT vs. CF Benchmarks USD), so their implied strikes disagree by the **USDT/USD stablecoin basis** — signed and forecastable before each hour resolves.
- Validated the mechanism on **1,350 real Kalshi settlements**: an offset-aware entry rule **removes the double-loss tail** (4.6% → 0%) and isolates the ~21% regime where both legs can win; characterized the two-sided cost distribution from live orderbook capture. Published it as a research note on SSRN, with an [interactive companion](#).
- Built a 24/7 execution and data-capture engine in **Rust** (chosen over Python for latency) on **AWS eu-west-1 (Ireland)** for low-latency dual-venue access; automated order placement, live orderbook capture, fill monitoring and **single-leg (“hanging-leg”) risk control**. Prototyped with two partners, then **rebuilt the entire system solo**.

Flight Dynamics Intern — PLD Space (launch-vehicle company, Elche, Spain) Mar – Apr & Jun – Aug 2025

- Built **MATLAB** tooling that automated recurring flight-dynamics analyses; reworked the **numerical surface-discretization scheme** behind the vehicle's aerodynamic force integration, improving its accuracy.

Co-founder & Sole Developer — Bolbee, AI marketing platform for SMEs 2025 – present

- Build and own the full product end-to-end (Next.js, Supabase/SQL, Stripe, multi-model AI pipeline) — architecture, data model, infrastructure and integrations; in private beta, market launch summer 2026.

Team Lead — Exoplanet Detection with ML, NASA Space Apps Challenge (Madrid) 2025

- Led a 3-person team building a neural-network exoplanet classifier on NASA telescope data (**Python**): owned data cleaning and feature selection, defined the train/test methodology and evaluation (confusion matrix); delivered a working model in 48 h.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas — research analysis & backtesting for the 1,350-settlement cross-venue study; data/eval pipeline for the NASA project), Rust (trading execution engine), MATLAB (flight dynamics, simulators), SQL, TypeScript.

Quantitative: convex optimization, probability & statistics, market microstructure, Gaussian processes, Monte Carlo methods, control & estimation theory, numerical optimization.

Tools: Linux (server administration), Git, AWS, REST APIs.

LANGUAGES

Spanish (native) | English (C1, Cambridge certified) | Catalan (C1)

HONORS & ADDITIONAL

- Merit scholarship: full first-year university tuition (ranked 1st in graduating high-school class).